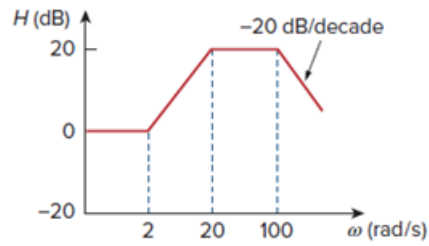


Problem

Find the transfer function $\mathbf{H}(\omega)$ with the Bode magnitude plot shown in Fig. 14.74.

Figure 14.74:



Step-by-step solution

Step 1 of 4

Refer to Figure 14.74 in the text book.

The magnitude at $\omega = 0.1 \text{ rad/s}$ is 0 dB.

Determine the value of K .

$$20 \log K = 0$$

$$\log K = 0$$

$$K = 1$$

Step 2 of 4

Determine the slope of the line between $\omega = 2$ and $\omega = 20$.

$$m = \frac{40 - 20}{\log 20 - \log 2}$$

$$= \frac{20}{\log \left(\frac{20}{2} \right)}$$

$$= \frac{20}{\log 10}$$

$$= 20 \text{ dB/decade}$$

The change in slope at $\omega = 2 \text{ rad/s}$ is,

$$\Delta m = 20 \text{ dB/decade} - 0 \text{ dB/decade}$$

$$= 20 \text{ dB/decade}$$

Obtain positive change in slope at $\omega = 2 \text{ rad/s}$ and therefore it is a zero $\left(1 + \frac{j\omega}{2}\right)^1$.

Step 3 of 4

The change in slope at $\omega = 20 \text{ rad/s}$ is,

$$\begin{aligned}\Delta m &= 0 \text{ dB/decade} - 20 \text{ dB/decade} \\ &= -20 \text{ dB/decade}\end{aligned}$$

Obtain negative change in slope at $\omega = 20 \text{ rad/s}$ and therefore it is a pole $\left(1 + \frac{j\omega}{20}\right)^{-1}$.

The change in slope at $\omega = 100 \text{ rad/s}$ is,

$$\begin{aligned}\Delta m &= -20 \text{ dB/decade} - 0 \text{ dB/decade} \\ &= -20 \text{ dB/decade}\end{aligned}$$

Obtain negative change in slope at $\omega = 100 \text{ rad/s}$ and therefore it is a pole $\left(1 + \frac{j\omega}{100}\right)^{-1}$.

Step 4 of 4

Determine the transfer function $H(\omega)$.

$$\begin{aligned}H(\omega) &= 1 \left(1 + \frac{j\omega}{2}\right)^1 \left(1 + \frac{j\omega}{20}\right)^{-1} \left(1 + \frac{j\omega}{100}\right)^{-1} \\ &= \frac{\left(1 + \frac{j\omega}{2}\right)}{\left(1 + \frac{j\omega}{20}\right) \left(1 + \frac{j\omega}{100}\right)}\end{aligned}$$

Thus, the transfer function $H(\omega)$ is

$$\boxed{\frac{\left(1 + \frac{j\omega}{2}\right)}{\left(1 + \frac{j\omega}{20}\right) \left(1 + \frac{j\omega}{100}\right)}}.$$